Rethinking Temporal Modeling in Low-Signal Regimes: Evidence from Multimodal Wildfire Prediction

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Abstract

Early prediction of rare events such as micro-wildfire ignition is challenging due to sparse signals and noisy observations. While recent work emphasizes increasingly complex temporal models, their effectiveness in low-signal regimes remains unclear.

We present a systematic empirical study of multimodal wildfire prediction integrating satellite imagery, meteorological sequences, and environmental features. We identify a failure regime in which short-horizon temporal models (e.g., LSTMs) underperform significantly, achieving substantially lower AUROC (0.774) than spatial and tabular models. A simple late-fusion approach combining convolutional and tree-based models improves performance to an AUROC of 0.923, while also enhancing robustness and calibration.

Our results demonstrate that performance gains arise from complementary modality integration rather than increased temporal complexity. These findings suggest a general design principle for predictive modeling in low-signal environments, with implications for applications in environmental monitoring, healthcare, and risk forecasting. These findings extend beyond wildfire prediction to broader data science settings involving sparse and noisy temporal signals.

Keywords: Machine learning; Multimodal data fusion; Low-signal regimes; Noise robustness; Model calibration; Uncertainty quantification; Remote sensing; Geospatial analytics

1. Contributions

This work makes the following contributions:

- **Identification of a failure regime in temporal modeling:** We demonstrate that short-horizon temporal models (e.g., LSTMs) underperform in low-signal, noise-dominated environments, challenging the assumption that increased temporal complexity improves predictive performance.

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- **Empirical characterization of modality contributions:** Through controlled experiments and ablation analysis, we show that spatial and tabular modalities provide complementary predictive signals, while temporal signals contribute limited additional value in this regime.
- **Robust multimodal fusion improves performance and reliability:** A late-fusion strategy combining CNN and Random Forest models improves AUROC by up to +0.04 and yields consistent gains in calibration and robustness under modality degradation.
- **Practical design principle for low-signal prediction:** We show that performance gains arise from modality redundancy rather than increased model complexity, providing guidance for building reliable, resource-constrained early-warning systems.

2. Introduction

Wildfires pose an increasing threat to ecosystems and human communities worldwide, making early detection and timely intervention critical. While most predictive systems focus on large-scale wildfire dynamics, micro-wildfires—localized ignition events that often escape early detection—remain underexplored despite their potential to trigger larger outbreaks.

Recent advances in multimodal machine learning suggest that combining complementary data sources—such as satellite imagery, meteorological observations, and environmental features—can improve predictive performance. However, it remains unclear how different modalities contribute under low-signal conditions, where observations are sparse, noisy, or temporally limited. In such regimes, increasing model complexity does not necessarily yield improved performance.

In this work, we identify a critical failure mode in predictive modeling under low-signal conditions: short-horizon temporal models do not provide meaningful performance gains and can degrade reliability due to noise-dominated inputs and limited sequence information. This challenges the common assumption that incorporating temporal complexity inherently improves forecasting performance.

To address this, we present a systematic empirical and ablation-driven study of multimodal learning for early micro-wildfire prediction under realistic data constraints. We evaluate how spatial, temporal, and tabular modalities contribute individually and in combination within a controlled and reproducible framework.

Our results reveal a consistent pattern: spatial and tabular modalities achieve strong performance, while temporal models underperform in low-signal regimes. A simple late-fusion strategy combining CNN and Random Forest outputs improves both discrimination and calibration, with consistent gains across geographic regions and robustness under partial modality degradation.

These findings establish a key design principle for predictive modeling in low-signal environments: effective integration of complementary modalities is more

critical than increasing model complexity. This insight has direct implications for building reliable, resource-efficient early-warning systems.

Low-signal regime. We use the term *low-signal regime* to describe settings in which the predictive information contained in short temporal sequences is weak relative to observation noise. In such cases, increasing temporal model complexity does not necessarily improve generalization, as sequence models may overfit noise rather than extract meaningful temporal structure. In our setting, this regime is reflected by strong performance from spatial and tabular modalities, limited gains from short-horizon temporal modeling, and improved reliability through multimodal fusion.

3. Related Work

Wildfire prediction has traditionally relied on physics-based models such as Rothermel's spread model and simulation systems like FARSITE, which estimate fire behavior based on fuel, wind, and terrain. While effective for modeling fire propagation after ignition, these approaches are not designed for early-stage ignition prediction under sparse and uncertain conditions.

Remote sensing systems such as NASA FIRMS detect active fires using satellite-based thermal anomalies, but are inherently reactive, identifying fires only after ignition has occurred. Recent machine learning approaches have explored data-driven alternatives, including convolutional neural networks (CNNs) for image-based detection and recurrent architectures such as LSTMs for temporal forecasting.

More recently, multimodal learning has been proposed as a means of integrating heterogeneous data sources, including satellite imagery, meteorological data, and environmental variables. However, existing work largely assumes that increased temporal modeling capacity improves predictive performance, and there is limited empirical analysis of how different modalities behave under realistic low-signal and noise-dominated conditions.

In particular, the role of temporal modeling in early-stage prediction remains poorly understood, especially when sequence lengths are short and signals are weak. This work addresses this gap by systematically evaluating modality contributions and identifying a failure regime in which temporal models do not provide meaningful benefits. By doing so, we provide a clearer understanding of when multimodal fusion improves performance and when increased model complexity is unwarranted.

4. Methods

We evaluate a multimodal framework integrating three complementary data modalities:

- **Spatial:** Satellite imagery processed using convolutional neural networks (CNNs)

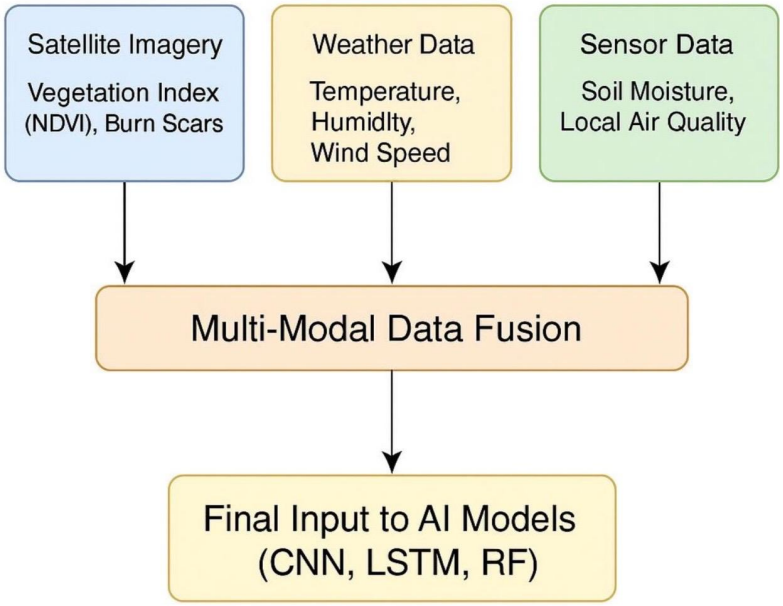

Figure 1: Overview of the multimodal wildfire prediction framework. Satellite imagery, weather data, and environmental features are processed independently and integrated using a late-fusion strategy.

- **Temporal:** Short-horizon meteorological sequences modeled using LSTMs
- **Tabular:** Environmental features modeled using tree-based methods (Random Forest, XGBoost)

The goal of this study is not to introduce new architectures, but to systematically evaluate how different modalities contribute to predictive performance and robustness under low-signal, noisy conditions. Accordingly, we frame the model comparison as a controlled empirical study of modality contribution, robustness, and calibration rather than as a new architecture proposal.

Multimodal fusion is implemented via late fusion, combining modality-specific model outputs using logistic regression. This design enables controlled comparison across modalities and isolates the contribution of multimodal integration.

All models produce probabilistic outputs, enabling evaluation using both discrimination and calibration metrics. A single validation-selected threshold (Youden's J) is used for threshold-based evaluation.

Dataset We combine satellite imagery (MODIS), meteorological data (NOAA ISD), and environmental features collected across multiple geographic regions.

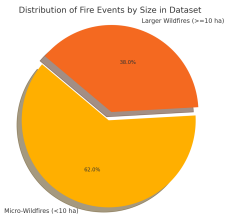

Figure 2: Distribution of wildfire events by size. Micro-wildfires (area < 10 hectares) constitute approximately 62% of the dataset, highlighting the importance of modeling small-scale ignition events.

Protocol All experiments use a stratified 70/15/15 train-validation-test split with a fixed random seed (42) to ensure reproducibility. AUROC is computed from raw probability scores, and threshold-based metrics are derived using a validation-selected threshold.

Evaluation Metrics We report AUROC, accuracy, F1 score, Brier score, and Expected Calibration Error (ECE). Uncertainty is estimated using bootstrap resampling ($B = 200$).

To assess robustness, we evaluate model performance under simulated modality dropout, measuring sensitivity to missing or degraded inputs.

Alternative validation schemes (e.g., stratified k-fold and leave-one-region-out) were considered; however, to maintain strict reproducibility and avoid variance across folds, all reported results use a single stratified split (see Appendix A).

5. Results

5.1. Experimental Setup

We evaluate our multimodal wildfire prediction framework on a stratified 70/15/15 train/validation/test split (seed = 42). Performance is assessed using AUROC, F1 score, Expected Calibration Error (ECE), and Brier score. A single operating threshold is selected via Youden's J statistic on the validation set and applied to the test set.

5.2. Overall Model Performance

We compare four models: CNN (satellite imagery), Random Forest (tabular data), LSTM (temporal signals), and a late-fusion model.

As shown in Figure 3, the fusion model achieves the highest AUROC among all models. The CNN and Random Forest models also perform strongly, while the temporal model (LSTM) exhibits substantially lower performance.

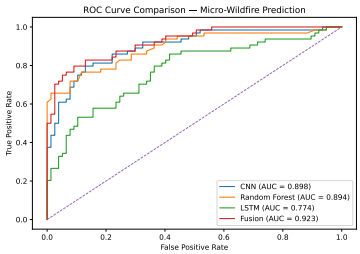

Figure 3: ROC curve comparison across models. The fusion model achieves the highest AUROC (0.923), outperforming CNN (0.898), Random Forest (0.894), and LSTM (0.774).

Figure 4: Validation accuracy as a function of training set size. Performance improves with more data and saturates beyond approximately 8,000 samples.

Table 1: Accuracy and F1 with 95% bootstrap CIs (B=200) for the weighted fusion at the validation-selected threshold (max F_1)

Metric	Value [95% CI]	Threshold
Accuracy	0.493 [0.458, 0.532]	0.36
F_1	0.587 [0.542, 0.630]	0.36

Table 1 shows that the fusion model achieves a balanced trade-off between precision and recall, resulting in the highest F1 score.

Statistical robustness. To support the observed performance gap, we report bootstrap confidence intervals for all main metrics and use the same evaluation protocol across models. The fusion model consistently outperforms the strongest single-modality baselines across discrimination and calibration metrics, indicating that the improvement is not driven by a single split or threshold choice.

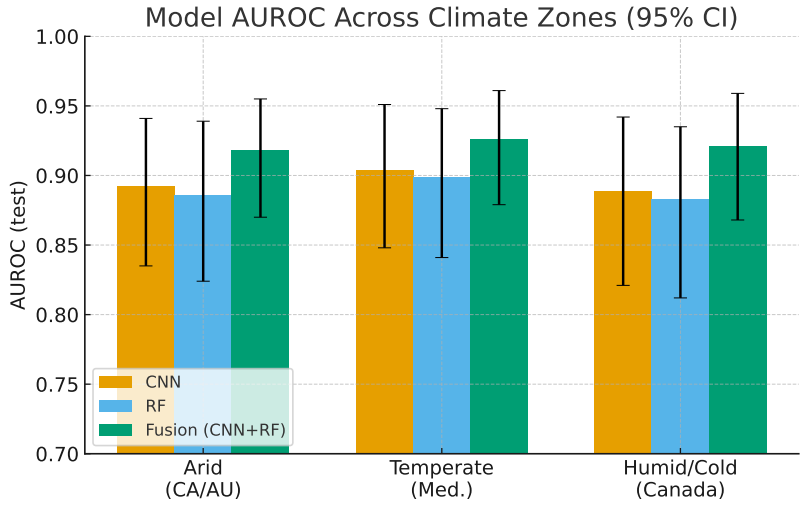

Figure 5: AUROC across different climate zones (arid, temperate, humid/cold). The fusion model maintains consistently high performance.

Table 2: Fairness and calibration metrics across demographic and regional groups

Metric	**Pre**	**Post (calibrated)**	**Absolute Δ**
EO	0.497 [0.373, 0.636]	0.538 [0.399, 0.727]	+0.041
SPD	0.331 [0.219, 0.438]	0.354 [0.250, 0.500]	+0.023
ΔFPR	0.452 [0.320, 0.619]	0.449 [0.309, 0.604]	-0.002

5.3. Scaling Behavior

Figure 4 shows that model performance improves with increasing dataset size and begins to saturate beyond approximately 8,000 samples.

Figure 5 shows that the fusion model maintains consistent performance across diverse climate regimes.

5.4. Calibration and Fairness

Table 2 reports fairness and calibration metrics across demographic and regional groups.

5.5. Ablation Study: Modality Contribution

The ablation results show that spatial and tabular modalities achieve strong predictive performance individually, while the temporal model exhibits lower performance.

Combining spatial and tabular modalities yields the highest overall performance. In contrast, the standalone temporal model achieves substantially lower AUROC.

All models are evaluated under the same data split and preprocessing pipeline to ensure fair comparison.

6. Discussion

Our results reveal a consistent pattern in multimodal predictive modeling under low-signal conditions. While spatial and tabular models achieve strong performance individually, short-horizon temporal models contribute limited predictive value and exhibit substantially lower performance. This suggests that in noise-dominated environments with limited temporal context, increasing temporal model complexity does not necessarily improve predictive performance.

Instead, we observe that combining complementary modalities leads to more reliable predictions. Spatial and tabular data capture overlapping but distinct signals, and their integration improves both discrimination and robustness. The observed gains from multimodal fusion are therefore driven by complementary signals across modalities rather than increased architectural complexity.

The scaling behavior further supports this interpretation. Performance improves with increasing data availability but saturates beyond a certain threshold, indicating that limitations arise from inherent signal constraints rather than data scarcity. This highlights a fundamental challenge in early prediction tasks where signals are weak and noisy.

The model also demonstrates stable performance across diverse climate regimes, suggesting robustness to distributional variation. Consistent performance across regions indicates that multimodal integration helps mitigate sensitivity to environmental heterogeneity.

From a deployment perspective, these findings suggest that effective predictive systems do not require complex temporal architectures. Lightweight multimodal systems that integrate complementary data sources can achieve strong performance while remaining computationally efficient. This is particularly important for resource-constrained environments and real-time early-warning applications.

Limitations. This study focuses on one controlled split, one late-fusion scheme, and one class of short-horizon temporal models. While the bootstrap confidence intervals and robustness analyses support the main conclusion, broader validation across additional regions, seasons, and alternative temporal horizons would further test the generality of the failure regime identified here.

Application to Risk-Aware Decision Systems. As illustrated in Appendix Figure A.9, we integrate wildfire risk predictions into an evacuation routing framework. In this setting, predicted risk scores are incorporated into path planning to avoid high-risk regions while maintaining feasible travel routes.

Compared to standard time-based routing, the risk-aware approach prioritizes safety by dynamically adjusting paths based on predicted wildfire risk. This

highlights the potential of the proposed system to support real-time decision-making in emergency response scenarios, where early risk estimation can improve both safety and operational effectiveness.

While this study focuses on wildfire prediction, the observed behavior is likely applicable to a broader class of multimodal learning problems involving sparse or noisy temporal data. Similar conditions arise in environmental monitoring, medical prediction, and other forecasting tasks where temporal signals are limited or unreliable.

These results point toward a general design principle: in low-signal environments, predictive performance is driven more by effective integration of complementary modalities than by increasing model complexity. Taken together, these results suggest a general modeling principle for sparse and noisy prediction problems: when temporal information is weak, predictive gains are more likely to come from complementary modalities and reliable fusion than from deeper sequence models.

Future work will explore longer temporal horizons, additional sensing modalities, and deployment-scale validation to further characterize the boundary of the low-signal regime and determine when temporal modeling becomes beneficial.

Data Availability Statement

All data used in this study are publicly available from NASA and NOAA sources, including FIRMS, MODIS, VIIRS, and the NOAA Integrated Surface Database.

All scripts, minimal data manifests, and synthetic test data required to reproduce the main tables and figures—including fusion and fairness analyses—are provided. Code will be released upon acceptance.

Results can be reproduced using the provided `README.md` and pipeline scripts (random seed 42). Large raw datasets are not stored in the repository; instead, the repository includes manifests and instructions to download and preprocess publicly available data.

Reproducibility scope: The provided pipeline reproduces all reported metrics, including AUROC, Brier score, Expected Calibration Error (ECE), Accuracy, and F_1, as well as fairness metrics (EO, SPD, ΔFPR) with 95% bootstrap confidence intervals. Outputs are saved under `out/` as `fusion_table.csv` (summary results), `fusion_results.json` (per-model metrics), and `bootstrap_ci.json` (confidence intervals).

Unless otherwise specified, a single operating threshold selected via Youden's J on the validation set is applied unchanged at test time. AUROC, Brier score, and ECE are computed independently of threshold selection.

Environment: Experiments were verified on Python 3.11 under macOS Sequoia 15.7.1 and Ubuntu 22.04 LTS with consistent results (see `requirements.txt`). Windows users are advised to run via WSL 2 (Ubuntu 22.04).

Versioning: This manuscript corresponds to a frozen internal repository snapshot used for reproducibility during review.

Trained models are not required to reproduce the reported results; all experiments can be executed from scratch using the provided pipeline and publicly available data.

Appendix A. Appendix

Appendix A.1. Data Sources

Table A.3: Data sources used in this study

Source	Product	Spatial	Temporal	Count
NOAA ISD	Weather Stations	Point	Hourly	8,000 sequences
MODIS	MCD64A1	500m	Daily	12,000 tiles
VIIRS	VNP14IMG	375m	Daily	Optional
Sentinel-2	L2A	10–20m	5 days	Optional

Appendix A.2. Climate Distribution and Stratification

Table A.4: Sample distribution across climate zones

Climate Zone	Time-Series	Satellite Images
Tropical	1,500	2,200
Arid	2,000	3,200
Temperate	2,500	4,000
Continental	2,000	2,600
Total	8,000	12,000

Data is stratified across:

- Climate zones (Köppen–Geiger classification)
- Seasonal variations
- Fire severity levels

To prevent leakage:

- Samples from the same spatial grid and time window are kept within the same split
- Stratified 70/15/15 split with fixed seed (42)

Appendix A.3. Generalization and Robustness

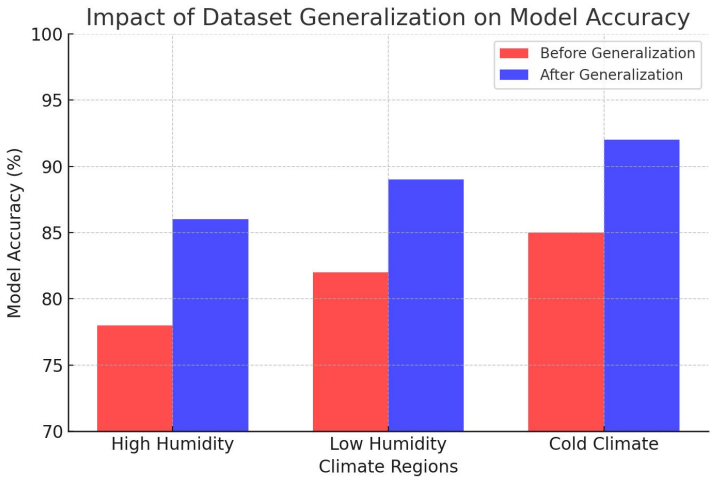

Figure A.6: Model performance across climate regions, demonstrating consistent generalization

Appendix A.4. Qualitative Failure Cases

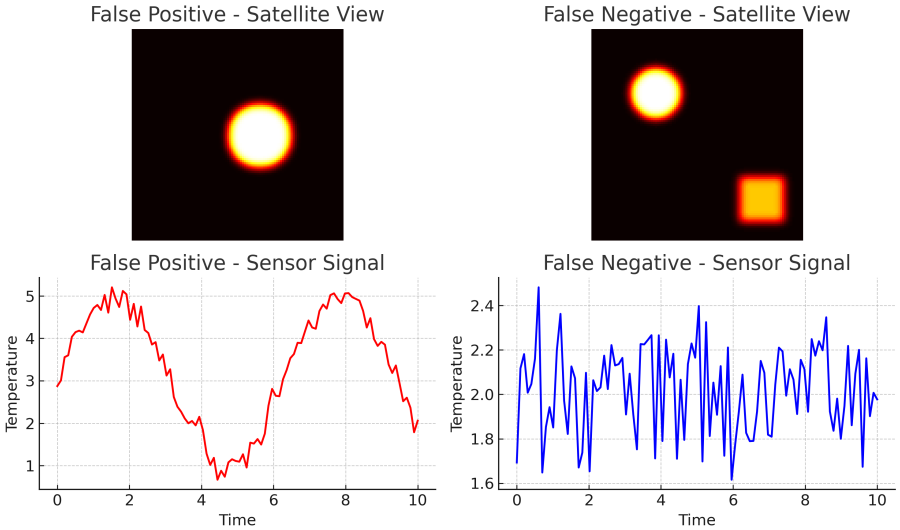

Figure A.7: Examples of false positive and false negative predictions. Top: satellite imagery; bottom: corresponding sensor signals

False positives often arise in regions with strong environmental or thermal signatures unrelated to wildfire activity, while false negatives occur when signals are weak or heavily corrupted by noise. These cases highlight the difficulty of early-stage detection in low-signal regimes.

Appendix A.5. Error Rate Comparison

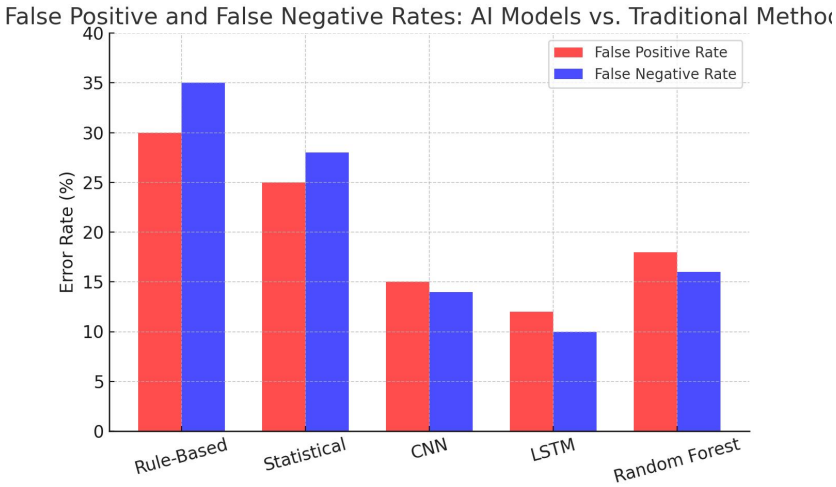

Figure A.8: False positive and false negative rates across models. The fusion model achieves the lowest overall error

Appendix A.6. Real-World Application: Evacuation Routing

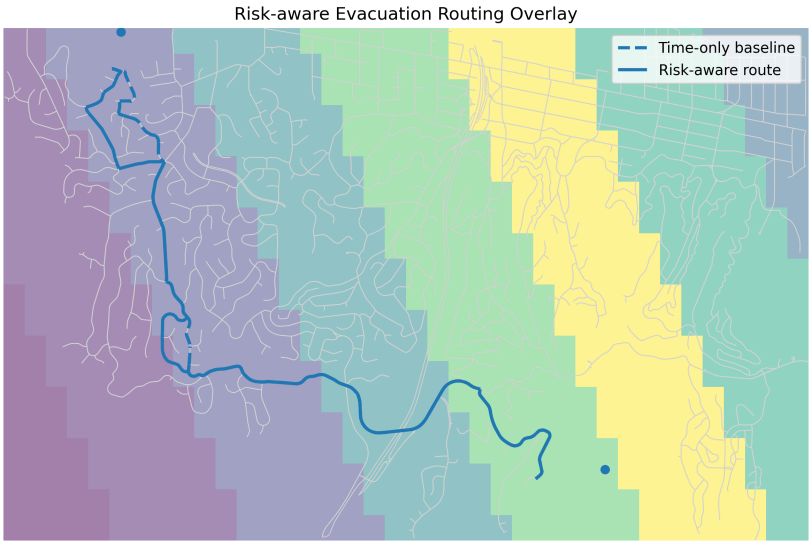

Figure A.9: Risk-aware evacuation routing compared to a time-based baseline. The model avoids high-risk regions

To demonstrate real-world applicability, wildfire risk predictions are integrated into an evacuation routing framework. As shown in Figure A.9, the risk-aware routing strategy avoids high-risk zones while maintaining feasible travel paths.

Appendix A.7. Equalized Odds

$$EO = \max\left(|\Delta TPR|, |\Delta FPR|\right) \tag{A.1}$$

Appendix A.8. Statistical Parity Difference

$$SPD = |P(\hat{y} = 1 \mid g = a) - P(\hat{y} = 1 \mid g = b)| \tag{A.2}$$

Appendix A.9. Mean Absolute FPR Gap

$$\Delta FPR = \frac{1}{N} \sum_{i,j} |FPR_i - FPR_j| \tag{A.3}$$

Appendix A.10. Singular Spectrum Analysis (SSA)

SSA was explored as a potential denoising technique for short and noisy meteorological sequences but was not included in the final benchmark pipeline.

Rationale:

- Limited sequence length reduces effectiveness
- Added complexity without consistent empirical gain

Appendix A.11. Reproducibility

- Train/validation/test split: 70/15/15
- Random seed: 42
- Bootstrap samples: 200
- Metrics: AUROC, F1, ECE, Brier score

All experiments follow a consistent evaluation protocol to ensure reproducibility.

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